

SAT Hard Question 13

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Question: Finding Circumference from Arc Length

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 60° , and the length of this arc is 4 inches.

What is the circumference, in inches, of the circle?

- A. 4
- B. 8
- C. 16
- D. 24

Answer Explanation: Finding Circumference from Arc Length (Part 1)

Step 1: The formula for arc length is:

$$\text{Arc Length} = \frac{\theta}{360^\circ} \cdot \text{Circumference}$$

where $\theta = 60^\circ$ and arc length is 4 inches. Let C be the circumference.

Answer Explanation: Finding Circumference from Arc Length (Part 1)

Step 1: The formula for arc length is:

$$\text{Arc Length} = \frac{\theta}{360^\circ} \cdot \text{Circumference}$$

where $\theta = 60^\circ$ and arc length is 4 inches. Let C be the circumference.

Step 2: Plug into the formula:

$$4 = \frac{60}{360} \cdot C \Rightarrow 4 = \frac{1}{6}C \Rightarrow C = 4 \cdot 6 = \boxed{24}$$

Correct Answer: (D) 24

Question: X-Intercepts of a Rational Function

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$.

The rational function g is defined by:

$$g(x) = \frac{f(x)}{x - 17}$$

In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

- A. 5
- B. 6
- C. 7
- D. 8

Answer Explanation: X-Intercepts of Rational Functions (Part 1)

Step 1: The original function $f(x)$ has 8 x-intercepts, including $x = 17$. This means $f(17) = 0$, so 17 is a zero of $f(x)$.

Answer Explanation: X-Intercepts of Rational Functions (Part 1)

Step 1: The original function $f(x)$ has 8 x-intercepts, including $x = 17$. This means $f(17) = 0$, so 17 is a zero of $f(x)$.

Step 2: In the function $g(x) = \frac{f(x)}{x-17}$, the zero at $x = 17$ is now in both the numerator and the denominator. This creates a removable discontinuity (a hole), not an x-intercept.

Step 3: The point $(17, 0)$ is no longer an x-intercept of $g(x)$. So, $g(x)$ has $8 - 1 = \boxed{7}$ x-intercepts.

Correct Answer: (C) 7

Question: Predator and Prey Growth Percent

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species.

At the end of the week:

- The number of prey had increased by 2700% of the number of prey at the start of the week.
- The number of predators had increased by 180% of the number of predators at the start of the week.

The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week.

What is the value of p ?

- A. 9
- B. 90
- C. 900
- D. 9000

Answer Explanation: Predator and Prey Growth Percent (Part 1)

Step 1: Assume initial populations.

Let the initial number of prey and predators be x .

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Step 2: Compute population at the end of the week.

- Prey: $x + 2700\% \cdot x = x + 27x = 28x$
- Predators: $x + 180\% \cdot x = x + 1.8x = 2.8x$

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- Predators: $x + 180\% \cdot x = x + 1.8x = 2.8x$

Step 3: Use percentage increase formula.

The prey population is $p\%$ greater than the predator population:

$$\frac{28x - 2.8x}{2.8x} = \frac{25.2x}{2.8x} = 9 \Rightarrow p = 900$$

Correct Answer: (C) 900

Question: Solving for Base of an Exponential Function

The exponential function f is defined by

$$f(x) = ab^x,$$

where a and b are positive constants.

If $f(s) = t$ and $f(s + 1) = t - 0.87t$, where s and t are constants, what is the value of b ?

- A. 0.13
- B. 0.87
- C. 1.13
- D. 1.87

Answer Explanation: Exponential Function (Part 1)

We are given:

$$f(s) = ab^s = t, \quad f(s+1) = ab^{s+1} = t - 0.87t = 0.13t$$

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We are given:

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Now divide the two expressions:

$$\frac{f(s+1)}{f(s)} = \frac{ab^{s+1}}{ab^s} = \frac{0.13t}{t} \Rightarrow \frac{b^{s+1}}{b^s} = b = 0.13$$

Correct answer: (A) 0.13

Question: Bee Population Prediction Model

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees.

The beekeeper uses a model that predicts the population of this bee colony:

- begins at 1,100
- increases by 120 bees per week in the first two weeks
- then increases by 180 bees per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

- A. $p(w) = 2,600 - 180w$
- B. $p(w) = 2,480 + 180w$
- C. $p(w) = 1,620 - 180w$
- D. $p(w) = -120 + 180w$

Answer Explanation: Bee Population Model

Step 1: Initial population is 1,100. **Step 2:** In first 2 weeks, increase is $2 \times 120 = 240$. So at the end of week 2, the population is:

$$1,100 + 240 = 1,340$$

Answer Explanation: Bee Population Model

Step 1: Initial population is 1,100. **Step 2:** In first 2 weeks, increase is $2 \times 120 = 240$. So at the end of week 2, the population is:

$$1,100 + 240 = 1,340$$

Step 3: For $w > 2$, increase is 180 bees per week. After w weeks, the population is:

$$\text{Population at week } w = 1,340 + 180(w - 2)$$

Step 4: Number of bees still needed to reach 2,600 is:

$$p(w) = 2,600 - (1,340 + 180(w - 2))$$

Answer Explanation: Bee Population Model

Step 1: Initial population is 1,100. **Step 2:** In first 2 weeks, increase is $2 \times 120 = 240$. So at the end of week 2, the population is:

$$1,100 + 240 = 1,340$$

Step 3: For $w > 2$, increase is 180 bees per week. After w weeks, the population is:

$$\text{Population at week } w = 1,340 + 180(w - 2)$$

Step 4: Number of bees still needed to reach 2,600 is:

$$p(w) = 2,600 - (1,340 + 180(w - 2))$$

Step 5: Simplify the expression:

$$\begin{aligned} p(w) &= 2,600 - 1,340 - 180(w - 2) = 1,260 - 180(w - 2) \\ &= 1,260 - 180w + 360 = 1,620 - 180w \end{aligned}$$

Correct answer: (C) $p(w) = 1,620 - 180w$

Question: Discriminant and Inequality in Quadratics

Given the quadratic equation:

$$7rx^2 - 28sx + 7r$$

In the equation:

- r is a positive constant
- s is a negative constant
- The equation has two solutions

If $\frac{r}{s} > k$, what is the minimum value of k ?

Answer Explanation: Discriminant and Inequality

Step 1: A quadratic equation has two distinct real solutions if its discriminant is positive:

$$\Delta = b^2 - 4ac > 0$$

Here:

$$a = 7r, \quad b = -28s, \quad c = 7r$$

Answer Explanation: Discriminant and Inequality

Step 1: A quadratic equation has two distinct real solutions if its discriminant is positive:

$$\Delta = b^2 - 4ac > 0$$

Here:

$$a = 7r, \quad b = -28s, \quad c = 7r$$

$$(-28s)^2 - 4(7r)(7r) > 0$$

$$784s^2 - 196r^2 > 0$$

Divide both sides by 196:

$$4s^2 - r^2 > 0 \Rightarrow 4s^2 > r^2 \Rightarrow \frac{r^2}{s^2} < 4 \Rightarrow \left| \frac{r}{s} \right| < 2$$

$$\frac{r}{s} > -2$$

Question: Map Scale Transformation

A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles.

A larger version of the same map is printed as a square with the side length 90% longer than the side length of the previous map.

On the larger map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- ☐ A 1.30
- ☐ B 6.84
- ☐ C 24.70
- ☐ D 49.50

Answer Explanation: Map Scale Transformation

Step 1: Original scale is:

$$\frac{13 \text{ miles}}{1 \text{ inch}} \text{ for a 55-inch map}$$

Step 2: New map side is $55 \times 1.9 = 104.5$ inches (90% longer).

Answer Explanation: Map Scale Transformation

Step 1: Original scale is:

$$\frac{13 \text{ miles}}{1 \text{ inch}} \text{ for a 55-inch map}$$

Step 2: New map side is $55 \times 1.9 = 104.5$ inches (90% longer).

Step 3: Since the physical map is larger, each inch on the larger map covers fewer miles. We apply a scale factor:

$$\text{Scale factor} = \frac{55}{104.5} \Rightarrow \text{New miles per inch} = 13 \times \frac{55}{104.5} \approx 6.84$$

Answer: 6.84

Question: Isosceles Triangle and Congruent Angles

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides side BC such that the length of $\overline{BD}$ is $\frac{4}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle $\angle BED$ is congruent to angle $\angle CFD$.

If the length of $\overline{BE}$ is 8, what is the length of $\overline{CF}$?

- (A) 6
- (B) 14
- (C) 32
- (D) 56

Answer Explanation: Isosceles Triangle and Congruent Angles

Step 1: Since $\angle BED \cong \angle CFD$, and triangles $\triangle BED \cong \triangle CFD$, then the triangles are similar by AA.

Answer Explanation: Isosceles Triangle and Congruent Angles

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Step 2: We are told that $BD : DC = 4 : 3$, since $BD = \frac{4}{7}BC$, meaning $DC = \frac{3}{7}BC$. This is the same as the ratio $\frac{BE}{CF}$.

$$\frac{BE}{CF} = \frac{4}{3} \Rightarrow \frac{8}{x} = \frac{4}{3} \Rightarrow x = \frac{8 \times 3}{4} = 6$$

Answer: 6

Question: Simplifying Radical Expressions

$$\frac{\sqrt[3]{x^{17}y^5}}{5x^2\left(\sqrt[8]{y^8}\right)}$$

For all positive values of x and y , the given expression is equivalent to which of the following?

1 $\frac{\left(x^{\frac{14}{3}}y^{\frac{5}{3}}\right)}{5xy}$

2 $\frac{\sqrt[3]{x^{11}y^2}}{5}$

A I only

B II only

C I and II

D Neither I nor II

Answer

Answer:

Linear Equation from Table — Parameterized

The table shows three values of x and their corresponding values of y , where s is a constant.

x	y
$-2s$	24
$-s$	21
s	15

There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $sx + 3y = 18s$
- B. $3x + sy = 18s$
- C. $3x + sy = 18$
- D. $sx + 3y = 18$

Answer: Linear Equation from Table

Use two known points: $x = -2s, y = 24$ $x = -s, y = 21$

Step 1: Find the slope

$$m = \frac{21 - 24}{-s - (-2s)} = \frac{-3}{s}$$

Step 2: Use point-slope form with point $(-s, 21)$

$$y - 21 = \frac{-3}{s}(x + s)$$

Step 3: Rearrange to standard form

$$sy - 21s = -3x - 3s \Rightarrow 3x + sy = 18s$$

Correct Answer: B. $3x + sy = 18s$

Function Modeling: Carpet Cleaner Rental Cost

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A. $C(d) = 26d + 26$

B. $C(d) = 26d + 52$

C. $C(d) = 52d - 26$

D. $C(d) = 52d + 78$

Answer: Carpet Cleaner Rental Function

Step 1: Understand the scenario

- The first day costs \$52.
- Each additional day costs \$26.

Step 2: Break the cost down

- If $d = 1$, the cost is \$52.
- If $d = 2$, the cost is $52 + 26 = 78$.
- For general d , think of:

$$C(d) = 52 + 26(d - 1)$$

- Simplify:

$$C(d) = 52 + 26d - 26 = 26d + 26$$

Correct Answer: A. $C(d) = 26d + 26$

Unit Conversion: Acceleration

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth?

(Use 1 mile = 1,609 meters)

- A. 0.3
- B. 16.3
- C. 195.8
- D. 220.4

Answer: Unit Conversion

Given:

$$7.3 \text{ m/s}^2$$

Convert to miles per minute squared:

$$\begin{aligned} & \frac{7.3 \text{ m}}{1 \text{ s}^2} \times \frac{1 \text{ mile}}{1,609 \text{ m}} \times (60 \text{ s})^2 \\ &= \frac{7.3}{1,609} \times 3,600 = 16.328 \approx \boxed{16.3} \end{aligned}$$

Correct Answer: B. 16.3